

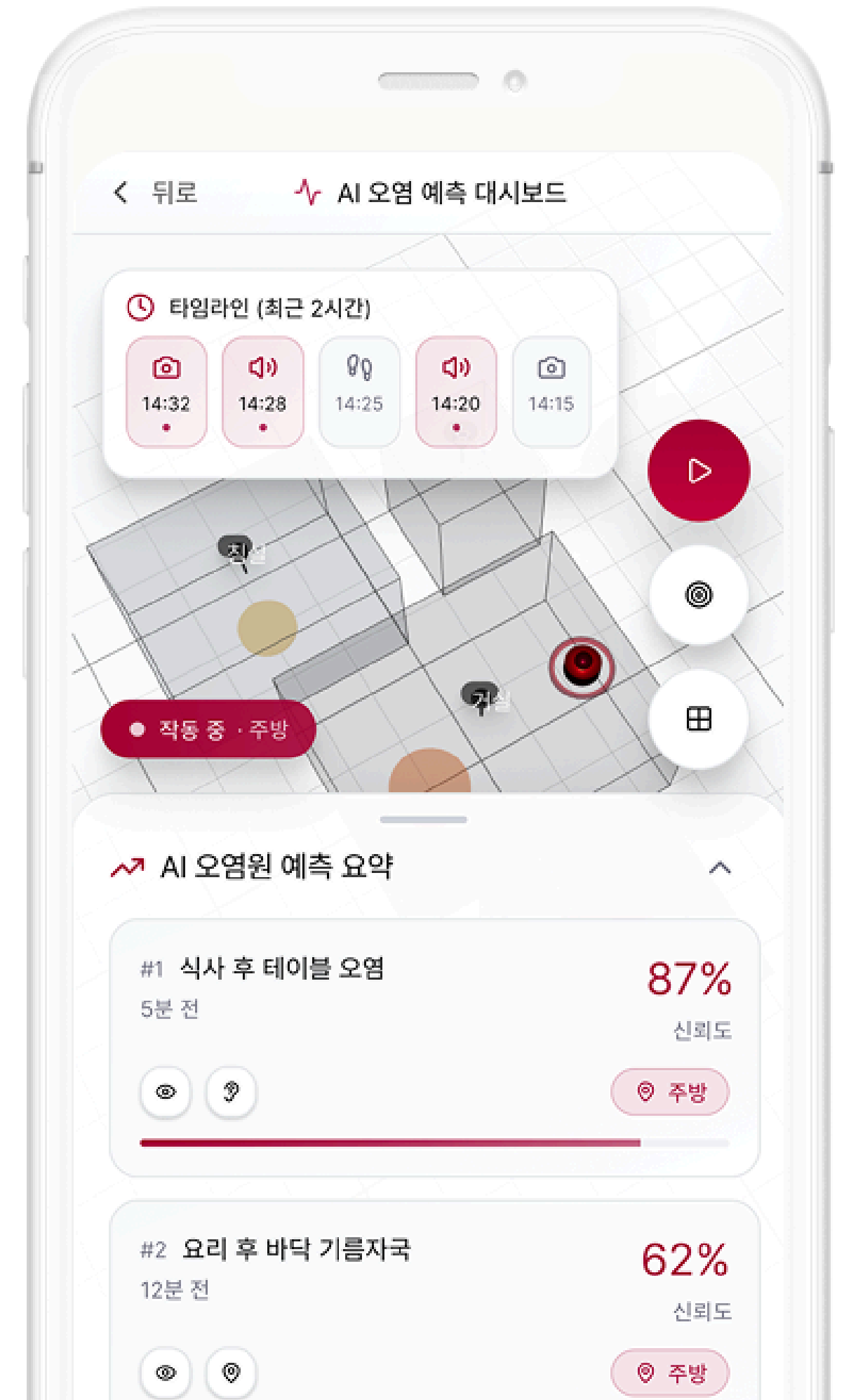
Software Engineering Project
2025 Hanyang University X LG Electronics



LOCUS

AI-Driven Household Context-Awareness
for Predictive Cleaning

고한영, 김준형, 이다연, 이승환



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Introduction



Motivation: The Limitations

Reactive Nature:

Current robots operate on preset schedules or direct commands.

Lack of Context:

Unable to respond to dynamic contamination or understand when & where.

Privacy Concern:

Conventional systems transmit sensitive raw data to cloud servers.

Q. What if we empowered the robot with Intelligence?



Our Goal : Active Partner

Predictive Cleaning:

Fuse visual, auditory, and positional info to predict cleaning needs.

On-device Privacy:

Process raw data locally; only share non-identifiable model weights via Federated Learning.

Personalization:

Learn user living patterns for optimized solutions.

What is LOCUS ?

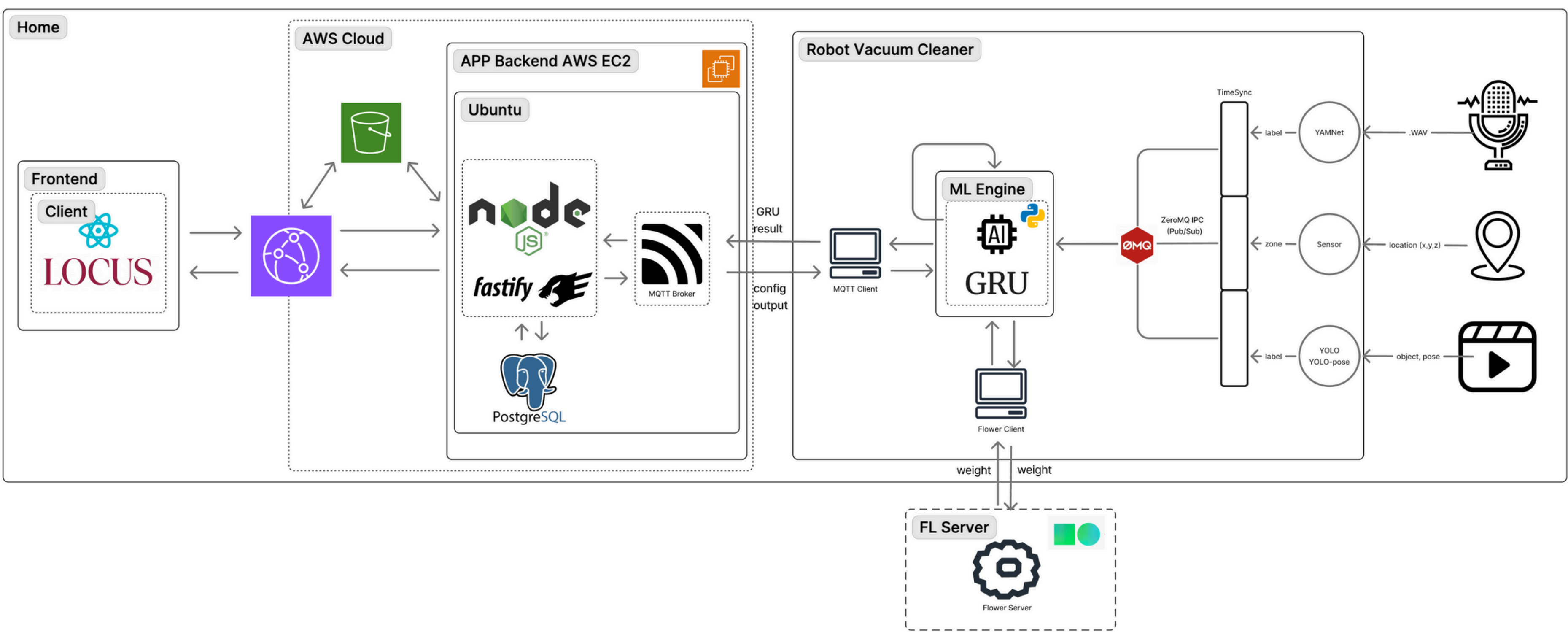
LOCUS

Learning On-device Context and User-specific Schedules

AI-Driven Household Context-Awareness for Predictive Cleaning



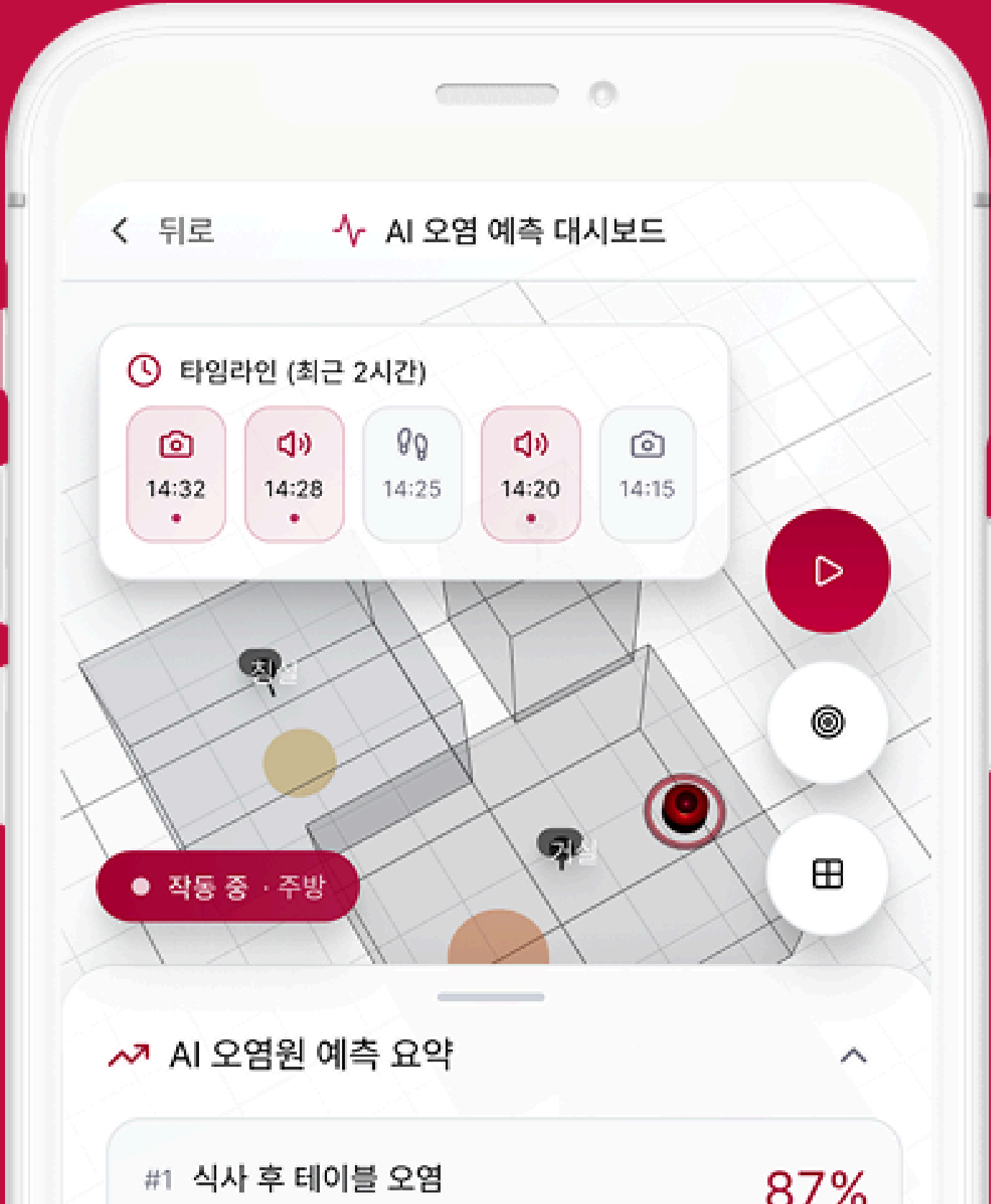
Architecture Design



FR1. Home Structure & Location Intelligence Module

Apple RoomPlan

ARKit



FR2. Multimodal Context Awareness



**Visual _ YOLOv11n
+ YOLOv11n-pose**

Dataset:
HomeObjects-3K + HD10K

Total : 8,689 Images

Classes: Bed, Sofa, Table,
Solid_waste, Liquid_stain, etc.



Auditory _ YAMNet

Dataset:
Refined Home Audio Dataset

Total : 22,212 Clips

Classes: Speech, Cooking, Dishes,
Water_tap, etc.

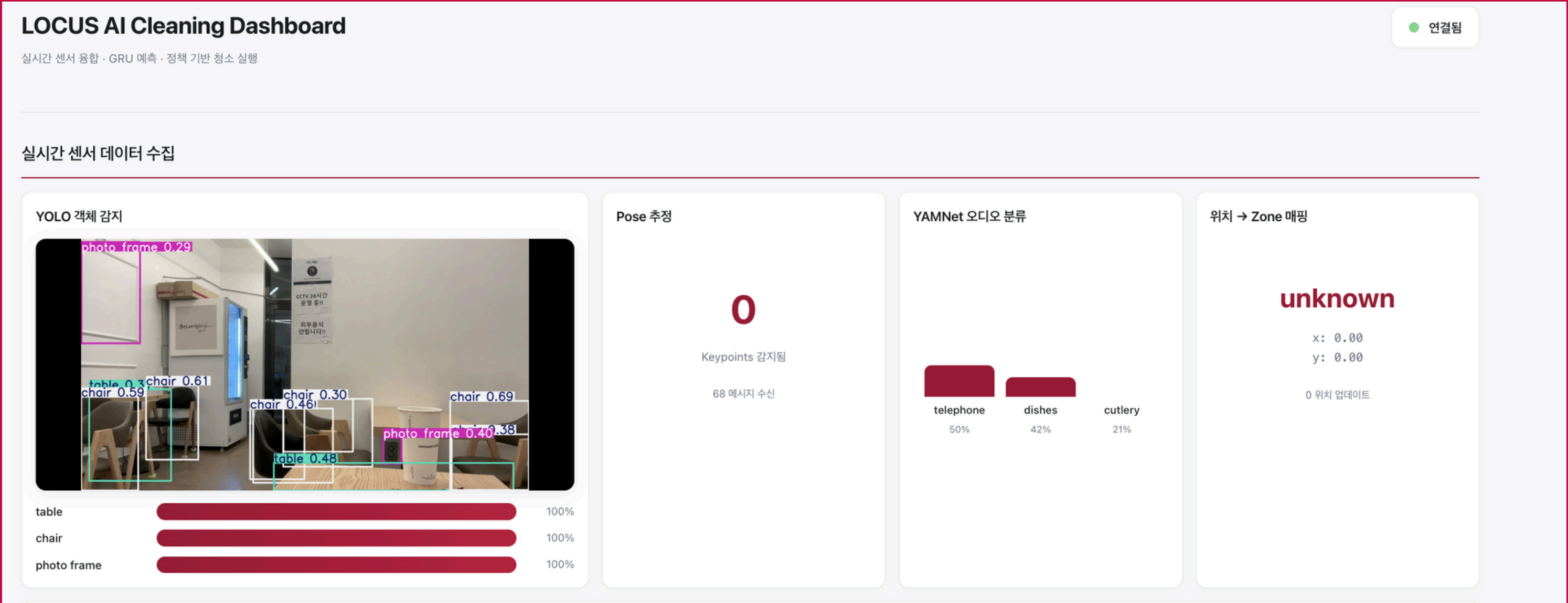


**Timestamp
Synchronization**

The system must synchronize
visual, audio, and location data
collected asynchronously.

Data is aligned within a $\pm 500\text{ms}$
window to generate a unified
context vector.

FR1 & FR2. Real-time Sensor Data Collection



Upon anomaly detection, the system synchronizes buffered sensor data to construct a 30-step sequence context for GRU inference.

FR3. Sequential GRU Prediction



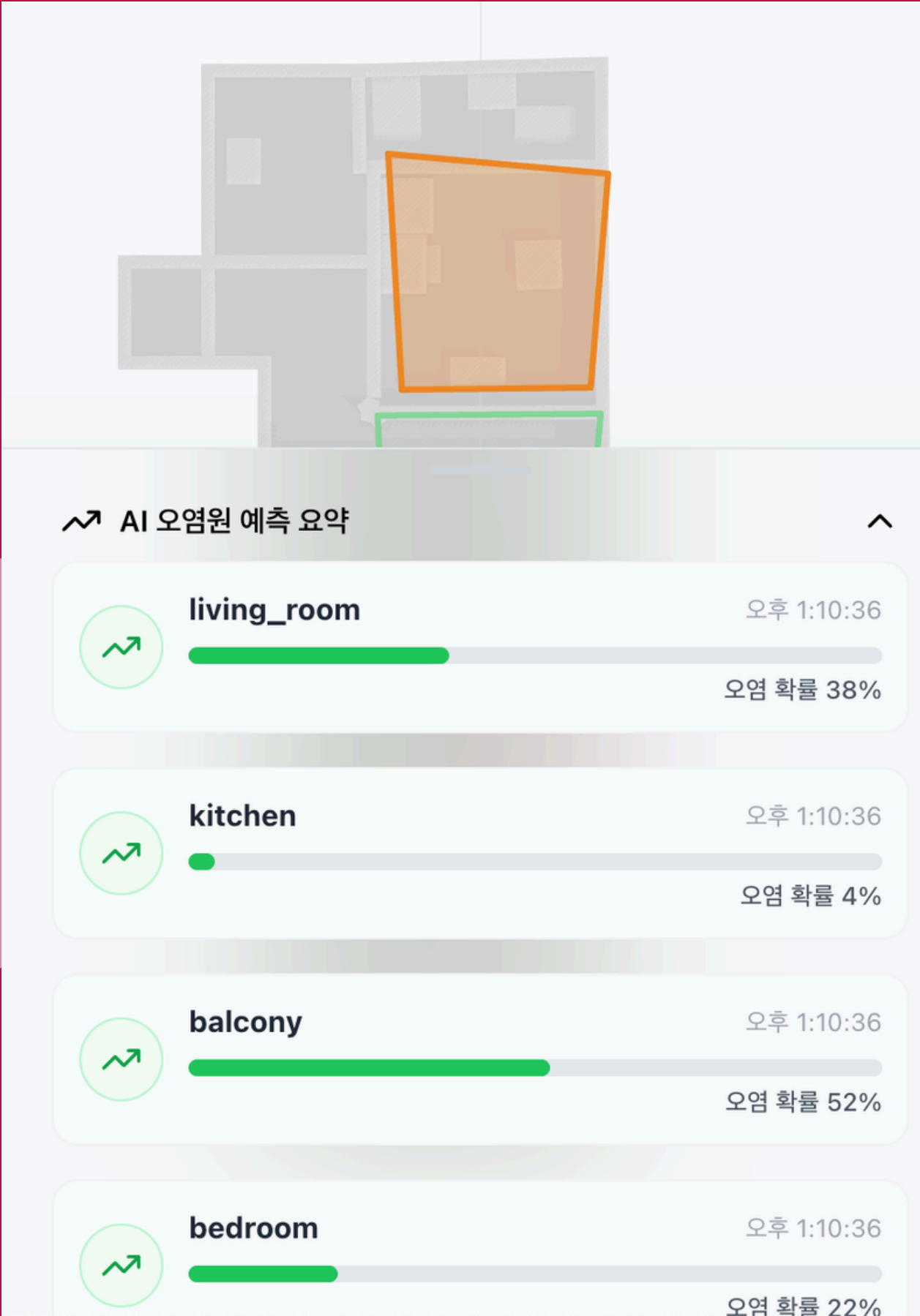
Context _ GRU

Dataset:
Multi-modal Context Vector

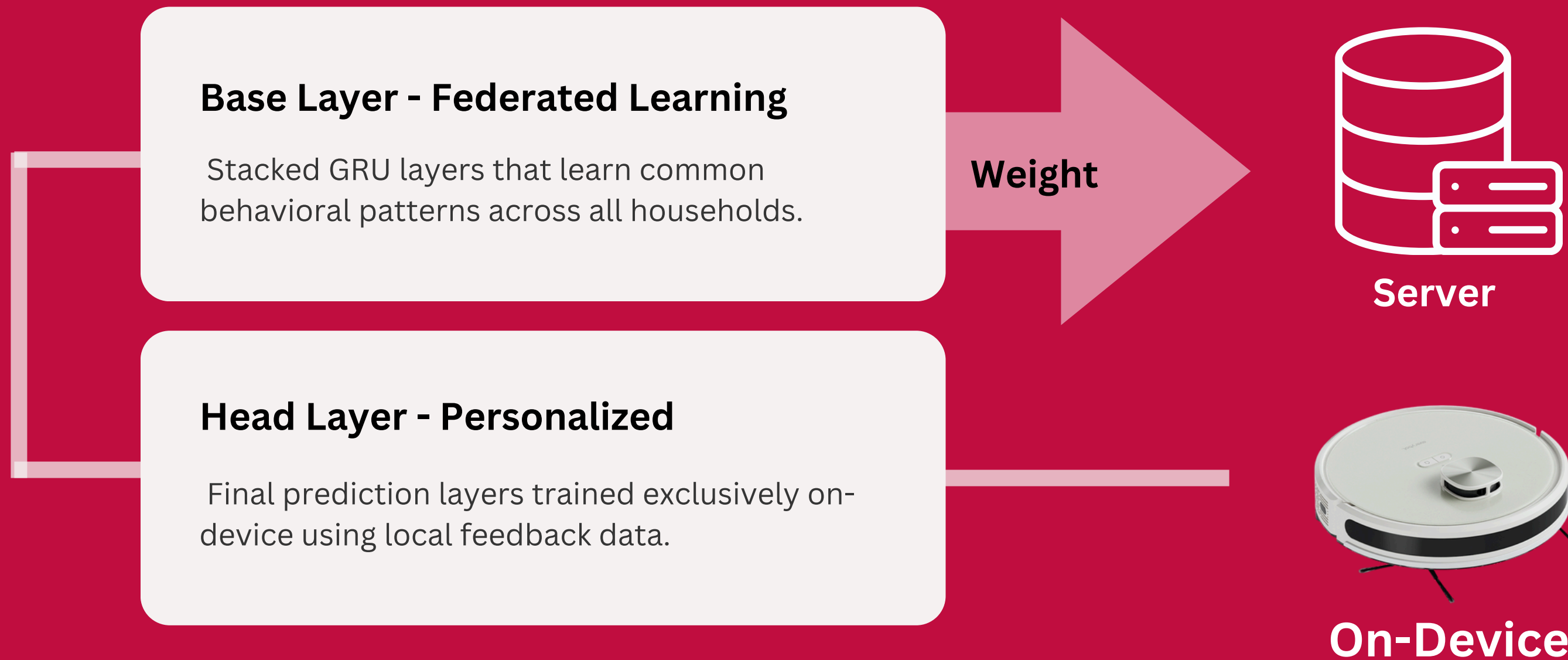
Input : 160-dim vector

Sequence Length: 30
Target: Zone prediction (Kitchen,
:Living Room, etc.)

Outputs
zone-specific
contamination
probability



FR4. Personalized Federated Learning



FR5. Cleaning Decision & Execution

FR5.1
Decision



Threshold Check:
Identify zones where
GRU prediction > 0.5

Priority Sorting:
Sort zones by
pollution probability
(descending)

Planning:
Plan cleaning sequence
starting from high-
pollution zones

FR5.2
Pre-verification

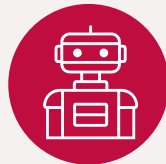


YOLO Detection:
IDetect solid waste
and liquid stains

Ground Truth:
Calculate actual
pollution score
weighted by object
type

Error Calculation:
Compute error
between predicted vs.
ground truth values

FR5.3 Execution
& Reporting



Operation:
Execute physical
cleaning tasks

Real-time Sync:
Transmit real-time
status via MQTT &
WebSocket

Completion:
Task completion and
switch to standby
mode

FR5.4
Feedback Loop



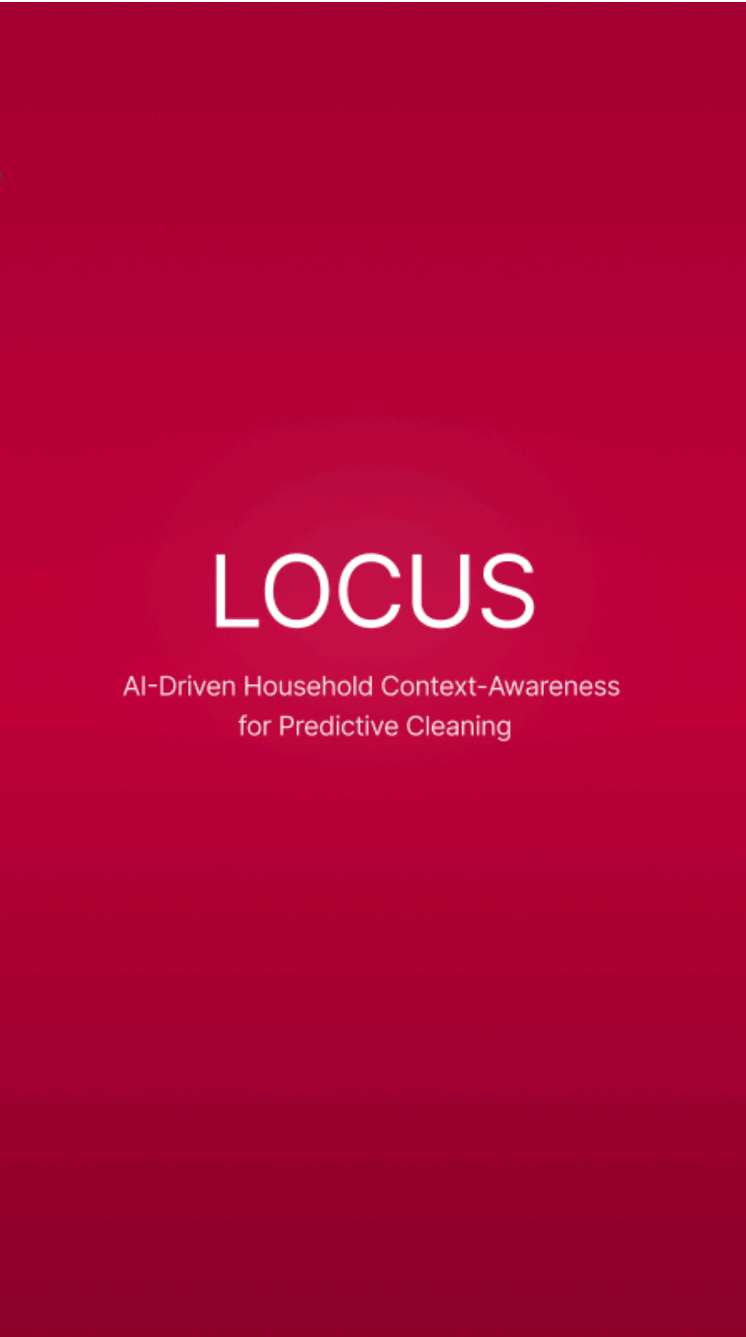
Data Collection:
Generate [GRU
Prediction - YOLO
Ground Truth] pairs

Auto-Training:
Trigger Head Layer re-
training upon data
buffer accumulation

Optimization:
Continuously optimize
prediction model
performance

Demonstration

Splash



Signup

9:41

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LG ThinQ와 함께하는
회원가입

이름

홍길동

이메일

example@email.com

중복확인

비밀번호

8자리 이상 입력

비밀번호 확인

비밀번호 재입력

☐ 전체 동의

☐ 이용약관 동의(필수) 보기

☐ 개인정보 처리방침 동의(필수) 보기

☐ 마케팅 정보 수신 동의(선택)

다음

이미 계정이 있으신가요? 로그인

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LOCUS

LG ThinQ와 함께하는 스마트 경험

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example@email.com

비밀번호

.....

비밀번호 찾기

로그인

또는 간편 로그인

Google로 계속하기

Apple로 계속하기

계정이 없으신가요? 회원가입

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Demonstration

Add Home

새로운 홈 시작하기

관리할 공간의 사진과 정보를 입력해주세요.

대표 사진 추가하기

터치하여 앨범에서 선택

홈 이름 (별명)

예: 우리집

주소 (선택)

예: 서울시 강남구

집 구조도

터치하여 .gltb 파일 업로드

* 3D Viewer에서 사용할 Room.gltb 파일을 업로드해주세요.

다음 >

9:41

홈 선택

관리할 홈을 선택하세요

+ 새로운 홈 추가

활성화

우리집

서울시 강남구

12개 기기 연결됨

활성화

부모님댁

경기도 성남시

8개 기기 연결됨

외출 중

별장

강원도 평창군

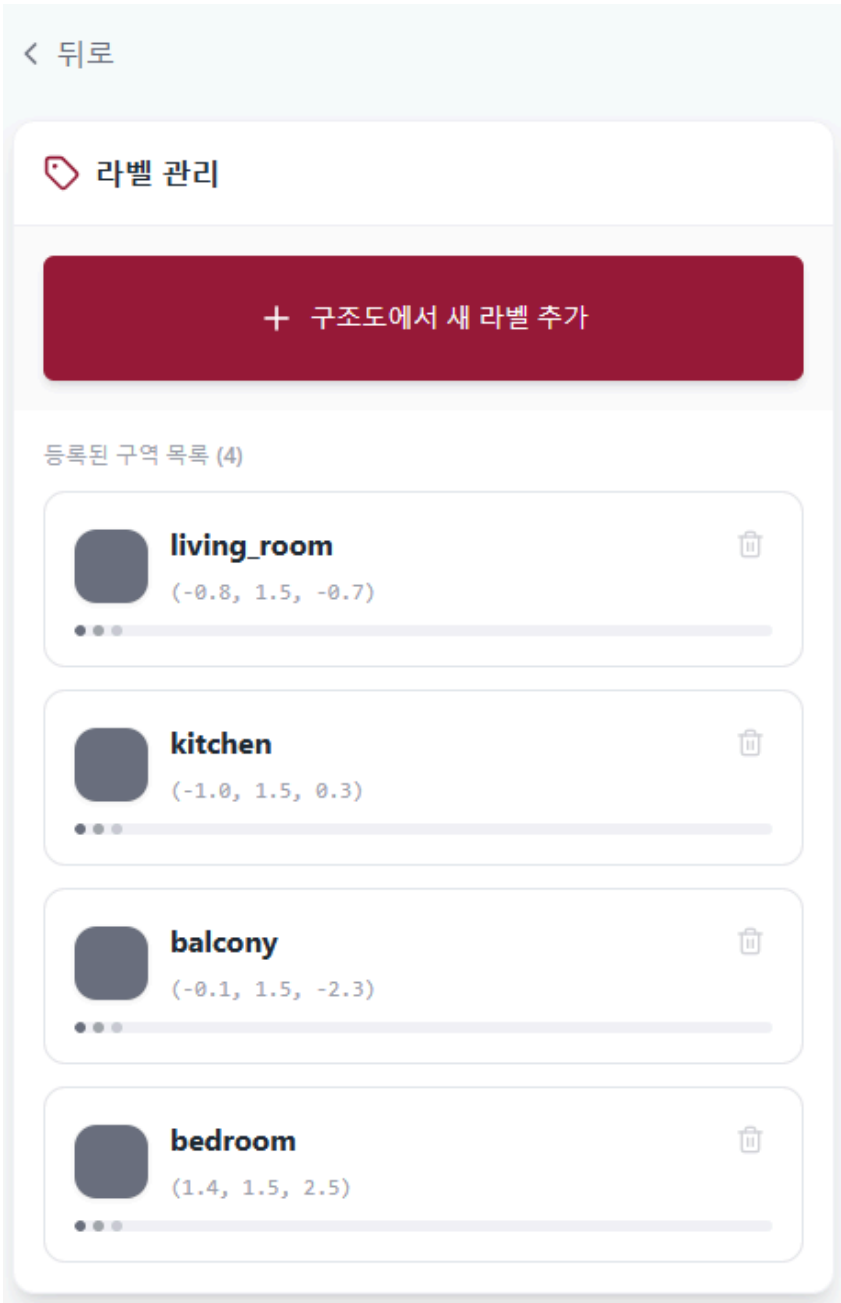
5개 기기 연결됨

여러 홈을 관리하세요

집, 사무실, 별장 등 여러 장소의 스마트 기기를 하나의 앱으로 편리하게 관리할 수 있습니다.

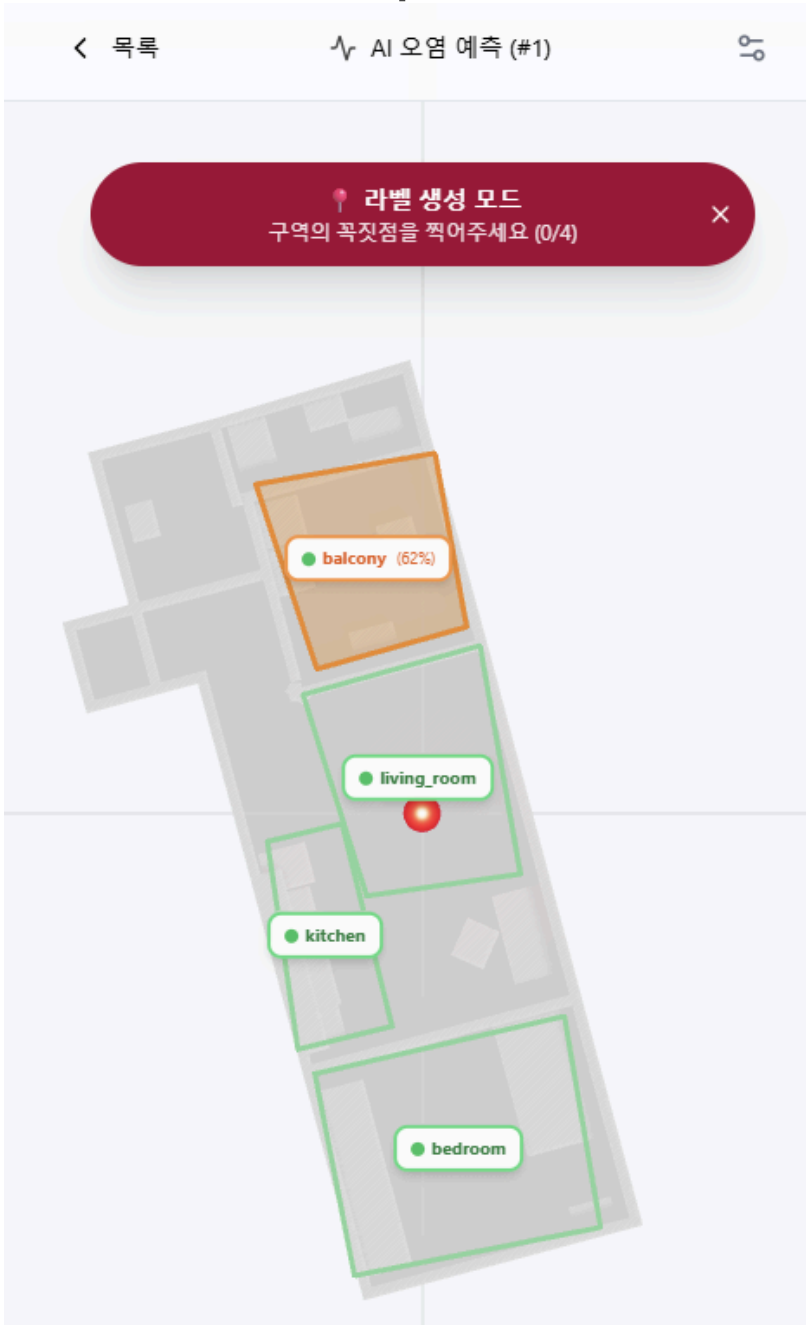
My Home

Demonstration



Label Management

Create Label



Delete Label

Demonstration



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Thank You

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X

